

1. Explain the notions of hypothesis testing, critical value, null distribution, alternative distribution, acceptance region, and rejection region. Use specific examples if necessary.
2. Consider the linear regression model: $Y_i = \beta_0 + \beta_1 X_i + u_i$, $i = 1, \dots, n$.
 - (a) Explain the distribution of the OLS estimator of β_1 in large samples.
 - (b) Suppose we want to test the hypothesis $H_0 : \beta_1 = 1$ vs. $H_1 : \beta_1 \neq 1$. Write down the formula of the t -statistic. Explain how we can obtain the large-sample distribution of this t -statistic using the result in (a).
3. Suppose that a random sample of size $n = 1,000$ is drawn from certain population and the observations (Y_i, X_i) , $i = 1, \dots, 1,000$ satisfy the OLS assumptions. The result of the OLS regression is:

$$\hat{Y}_i = 1.5 + 2.4X_i, i = 1, \dots, 1000, \quad (1)$$

with the standard error of $\hat{\beta}_0$ equal to 1.0 and the standard error of $\hat{\beta}_1$ equal to 1.2.

- (a) Test the slope of the regression using $H_0 : \beta_1 = 0$ vs $H_1 : \beta_1 \neq 0$ at the 5% significance level. Similarly, test the intercept of the regression at the 5% significance level.
 - (b) Suppose that X_i actually has no effect on Y_i . Also suppose that 2,000 random samples of size $n = 1,000$ are drawn repeatedly from the population, and the OLS regressions are estimated repeatedly 2,000 times (suppose the sampling process always satisfies the OLS assumptions). In how many of these random samples would the null hypothesis $H_0 : \beta_1 = 0$ be rejected by the (two-sided) t -test at the 10% significance level? Explain.
4. (a) Consider the following linear regression model for saving and income:

$$saving_i = \beta_0 + \beta_1 income_i + u_i, u_i = \sqrt{income_i} \cdot e_i,$$

where e_i satisfies $E[e_i] = 0$ and $Var[e_i] = \sigma^2$. In addition, e_i is independent of $income_i$. First, show that the conditional expectation of e_i given the value of $income_i$ is equal to zero (by using the property of independence between two random variables). Second, show that this model satisfies the OLS Assumption 1. [Hint: Notice that the following result holds for random variables X and Y for any function $g(X)$: $E(g(X) \cdot Y|X) = g(X) \cdot E(Y|X)$.]

- (b) Consider the linear regression model $Y_i = \beta_0 + \beta_1 X_i + u_i$. Suppose you know that $\beta_0 = 0$. Derive a closed-form formula for the OLS estimator of β_1 . [Hint: As we know $\beta_0 = 0$, there is no need to estimate it in this case. Therefore, the minimization problem becomes: $\min_{b_1} \sum_{i=1}^n (Y_i - 0 - b_1 X_i)^2$, i.e., $\min_{b_1} \sum_{i=1}^n (Y_i - b_1 X_i)^2$.]